

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 2, Multicenter, Open-Label Study to Assess Appropriate Dosing and to Evaluate Safety of Crizanlizumab, With or Without Hydroxyurea/Hydroxycarbamide, in Sequential, Descending Age Groups of Pediatric Sickle Cell Disease Patients With Vaso-Occlusive Crisis
Short Title:	Study of Dose Confirmation and Safety of Crizanlizumab in Pediatric Sickle Cell Disease Patients
Protocol Number:	739628149306423520
NCT Number:	NCT03474965
Sponsor Name:	Novartis
Investigational Product:	crizanlizumab
Phase:	PHASE2
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Pharmacokinetics (PK): AUC _{d15} of Crizanlizumab After First Dose - Part A: The area under the curve (AUC) from time zero to the last measurable concentration sampling time (t _{last}) (mass x time x volume ⁻¹) following the first dose. AUC _{d15} was calculated based on serum concentrations of crizanlizumab. 2. Pharmacokinetics (PK) - AUC _{tau} for Serum Crizanlizumab After Multiple Doses - Part A - Steady State: The AUC calculated to the end of a dosing interval (tau) at steady-state (amount x time x volume ⁻¹). 3. Pharmacokinetics (PK) - C _{max} for Crizanlizumab After First Dose and Multiple Doses - Part A - Steady State: The maximum (peak) observed, serum, drug concentration after single or multiple dose administration (mass x volume ⁻¹)
Secondary Obj.:	1. Annualized Rate Vaso Occlusive Crisis (VOC) Events Leading to Healthcare Visit in Clinic / Emergency Room (ER) / Hospital 2. Annualized Rate Vaso Occlusive Crisis (VOC) Events Treated at Home (Based on Documentation by Health Care Provider Following Phone Contact With the Patient) 3. Annualized Rate Vaso Occlusive Crisis (VOC) Events Leading to Healthcare Visit - Uncomplicated Pain Crisis

2.2 Background & Rationale

The purpose of this study was to confirm and to establish appropriate dosing and to evaluate the safety in pediatric participants ages 2 to <18 years with a history of Vaso-Occlusive Crisis (VOC) with or without Hydroxyurea/Hydroxycarbamide (HU/HC), receiving crizanlizumab for 2 years. The efficacy and safety of crizanlizumab was previously demonstrated in adults with sickle cell disease. The approach was to extrapolate from the pharmacokinetics (PK)/pharmacodynamics (PD) already established in the adult population. The study was designed as a Phase II, multicenter, open-label study.

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3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE2, Status: COMPLETED

Target N: 117

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

1. Male or female patients ages 2 to <18 years
2. Confirmed diagnosis of SCD (any genotype including HbSS, HbSC, HbS β 0-thalassemia, HbS β +thalassemia patients, and others) by hemoglobin electrophoresis or/and high-performance liquid chromatography (HPLC) [performed locally]. Confirmation of diagnosis by two accepted methods is recommended.
3. Experienced at least 1 VOC within the preceding 12 months prior to screening, as determined by medical history. Prior VOC must have resolved at least 7 days prior to the first dose in the study and must include all the following: a.the occurrence of appropriate symptoms (see VOC definition in Section 7.2.1.1), b.either a visit to a medical facility or healthcare professional, c.receipt of oral/parenteral opioid or parenteral NSAIDs
4. If receiving HU/HC, L-glutamine or erythropoietin stimulating agent, must have been receiving the drug consistently for at least 6 months prior to screening and plan to continue taking it at the same dose and schedule during the trial. Patients who have not been receiving such drugs must have been off them for at least 6 months prior to screening. . Dose alterations of HU/HC, L-glutamine or erythropoietin stimulating agent during Part A are not allowed, and if this occurs, the participant will enter directly to Part B.
5. Received standard age-appropriate care for SCD, including penicillin prophylaxis, pneumococcal immunization, and parental education.
6. Performance status: Karnofsky $\geq$ 50% for patients $\geq$ 10 years of age, and Lansky $\geq$ 50 for patients $\geq$ 10 years of age.
7. Patient must meet the following laboratory values prior to Week 1 Day 1: Absolute Neutrophil Count $\geq$ 1.0 x 10⁹/L , Platelets $\geq$ 75 x 10⁹/L, Hemoglobin (Hgb) $\geq$ 5.5 g/dL
8. Patient must have adequate renal and hepatic function as defined:Estimated Glomerular filtration rate (eGFR) $\geq$ 75 mL/min/1.73 m² using Schwartz formula, Direct (conjugated) bilirubin $\leq$ 2.0 x ULN, Alanine transaminase (ALT) $\leq$ 3.0 x ULN,
9. Transcranial Doppler (TCD) for patients aged 2 to < 16 years at time of screening, with HbSS, HbS β 0-thalassemia, and HbSD disease indicating low risk for stroke (per investigator). Please refer to Section 7.2.2.6 for details
10. Written informed consent/assent, according to local guidelines, signed by the patient and / or by the parents or legal guardian prior to any study related screening procedures are performed.
11. Female of non-childbearing potential or with negative serum pregnancy test on Screening and a negative urine pregnancy test (dipstick) prior to dosing on Day 1.

4.2 Exclusion Criteria

1. History of stem cell transplant.
2. Received any blood products within 30 days prior to Week 1 Day 1 dosing.
3. Plan to participate in a chronic transfusion program (pre-planned series of transfusions for prophylactic purposes) or undergo exchange transfusions/plasmapheresis during the study. Patients requiring episodic transfusion (simple or exchange) in response to worsened anemia or VOC are permitted.

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4. Patients with bleeding disorders

6. Contraindication or hypersensitivity to any drug from similar class as study drug or to any excipients of the study drug formulation.

7. History of severe hypersensitivity reaction to other monoclonal antibodies, which in the opinion of the investigator may pose an increased risk of serious infusion reaction 8. Received a monoclonal antibody or immunoglobulin-based therapy within 6 months of Screening, or has documented immunogenicity to a prior monoclonal antibody.

9. Received active treatment on another investigational trial within 30 days (or 5 half-lives of that agent, whichever is greater) prior to Screening or plans to participate in another investigational drug trial.

10. Pregnant females or females who have given birth within the past 90 days or who are breastfeeding.

11. Any documented history of a clinical stroke or intracranial hemorrhage, or an uninvestigated neurologic finding within the past 12 months. Silent infarcts (only present on imaging) are not excluding patients from study participation

12. Any abnormal TCD within the past 12 months. 13. Use of therapeutic anticoagulation (prophylactic doses permitted) or antiplatelet therapy (other than aspirin) within the 10 days prior to Week 1 Day 1 dosing.

14. Hospitalized within 7 days prior to Week 1 Day 1 dosing. 15. Planning to undergo a major surgical procedure during the duration of the study.

16. Planning to initiate or terminate HU/HC or L-glutamine while on study (except if needed to terminate for safety reasons).

17. Patient with active human immunodeficiency virus (HIV) infection (detectable viral load).

18. Patients with known active Hepatitis B infection. 19. Patients with known Hepatitis C history. 20. Significant active infection or immune deficiency (including chronic use of immunosuppressive drugs) in the opinion of the investigator.

21. Malignant disease. Exceptions to this exclusion include the following: malignancies that were treated curatively and have not recurred within 2 years prior to study treatment; any completely resected carcinoma in situ.

22. Has a serious mental or physical illness, which, in the opinion of the Investigator would compromise participation in the study.

23. Any condition which, in the opinion of the investigator, is likely to interfere with the successful collection of the measurements required for the study 24. Resting QTcF ≥ 450 msec at pretreatment (baseline) for patients under 12 years of age and ≥ 450 msec for males and ≥ 460 msec for female patients 12 years and older.

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25. Cardiac or cardiac repolarization abnormality, including any of the following: a. History of myocardial infarction (MI), uncontrolled congestive heart failure, unstable angina, or coronary bypass graft (CABG) within 6 months prior to starting study treatment, b. Clinically significant cardiac arrhythmias (e.g., ventricular tachycardia), complete left bundle branch block, high-grade AV block (bifascicular block, Mobitz Type II, and third degree AV block), c. Long QT syndrome, family history of idiopathic sudden death or congenital long QT syndrome (Risk factors for Torsade de Pointes (TdP), including uncorrected hypokalemia or hypomagnesemia, history of cardiac failure, or history of clinically significant/ symptomatic bradycardia, Inability to determine the QTcF).

26\ Sexually active females who are unwilling to comply with reliable method of birth control until 15 weeks following last dose of study drug.

28. Not able to understand and to comply with study instructions and requirements.

29. Patients who are an employee of the sponsor or investigator or otherwise dependent on them.

30. Patients who are committed to an institution by virtue of an order issued either by the judicial or the administrative authorities.

31. Patients who received prior crizanlizumab treatment and/or other selectin targeting agents are not allowed

32. Patients having taken voxelotor less than 30 days prior to Screening, or planning to take voxelotor while on study are not allowed.

11. Study Identifiers & Governance

Primary ID:	739628149306423520
Registry Number:	NCT03474965
Sponsor:	Novartis
Study Title:	Study of Dose Confirmation and Safety of Crizanlizumab in Pediatric Sickle Cell Disease Patients

15. Sign-off & Approval

Principal Investigator: _____	Date: _____
Clinical Operations Lead: _____	Date: _____
Lead Statistician: _____	Date: _____